



Drowning / Submersion

Assessment

Pediatric Pearls:

- Use pediatric dosing of medications or electrical therapy for a pediatric patient < 37 kg and as defined by the PEDIA tape.
- Pediatric hypotension is defined as SBP < 70 + (age in years x 2) mmHg

Signs & Symptoms:

- Unresponsive
- Mental status changes
- Decreased or absent vital signs
- Vomiting
- Coughing

Differential:

- Trauma
- Pre-existing medical problem
- Pressure injury (diving)
 - Barotrauma
 - Decompression sickness
- Duration of immersion
- Temperature of water

Clinical Management Options

P	P	P	P	P	P
L	L	L	L	L	L
1	2	3	4	5	6

- Scene safety & decontaminate patient as needed
- Evaluate need for [spinal motion restriction](#)
- [Oxygen](#), Target SpO₂: 92-94%
- BLS airway management as needed, consider [PEEP](#) (5-10mmHg)
- Suction
- Keep patient warm
- If water temperature <10°C/<50°F, [Hypothermia COG](#)
- If conscious and with wheezing, [Albuterol](#) & [Ipratropium Bromide nebulizer](#)
- If respiratory distress, [CPAP](#)
- ECG placement
- [Vascular access](#)
- Evaluate ECG
- Advance airway maneuvers and [management](#) as needed

Consult Online Medical Control As Needed

Pearls:

- Refer to drug formulary charts for all medication dosing for both adults and pediatric patients.
- Consider withholding resuscitation or conversion to recovery efforts if:
 - Clear history of prolonged submersion >20 min unless patient is a diver with air source
 - > 20min from arrival of first public safety entity to when patient is in a position for resuscitation
- **IF water temperature <6°C(43°F)**, consider continuing rescue/resuscitation efforts for 90 minutes – refer to [hypothermia COG](#)

Final decision for transition from rescue to recovery rests with on-scene command

- SMR should be used when a suspected or known traumatic mechanism preceded the drowning.
- All victims should be transported for evaluation due to potential for worsening over the next several hours.
- With pressure injuries (decompression/barotrauma), transport dive computer and/or dive logs with patient if possible.
- Consider CPAP early
- Be prepared for copious airway secretions
- Supraglottic airways may not be able to handle high PEEP; intubate if necessary